

Matematica - Clasa a 5-a

Exercitii: Capitol IV - Numere rationale

Nume: _____

Total: 100 puncte + 10 puncte din oficiu

Data: _____

Ex. 1. Incercuieste ADEVARAT sau FALS:

(10p)

- Orice numar natural este si rational ADEVARAT / FALS
- $\frac{3}{4}$ este o fractie supraunitara ADEVARAT / FALS
- $\frac{6}{9}$ este ireductibila ADEVARAT / FALS
- $\frac{1}{3} = \frac{2}{6} = \frac{4}{12}$ ADEVARAT / FALS
- $0,25 = \frac{25}{100} = \frac{1}{4}$ ADEVARAT / FALS
- $\frac{3}{5} > \frac{4}{7}$ (prin produse incrucisate: $3 \times 7 = 21 > 5 \times 4 = 20$) ADEVARAT / FALS
- $\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$ ADEVARAT / FALS
- $\frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$ ADEVARAT / FALS
- $\frac{3}{4} : \frac{3}{8} = 2$ ADEVARAT / FALS
- $1,2 \times 3 = 3,6$ ADEVARAT / FALS

Ex. 2. Fractii egale si ireductibile:

(10p)

a) Reduce la forma ireductibila (calculeaza c.m.m.d.c. si simplifica):

$$\frac{24}{36} = \frac{\quad}{\quad}$$

$$\frac{45}{60} = \frac{\quad}{\quad}$$

$$\frac{84}{126} = \frac{\quad}{\quad}$$

$$\frac{120}{180} = \frac{\quad}{\quad}$$

$$\frac{72}{96} = \frac{\quad}{\quad}$$

$$\frac{210}{315} = \frac{\quad}{\quad}$$

b) Completeaza pentru a obtine fractii egale:

$$\frac{3}{4} = \frac{\quad}{20} = \frac{\quad}{100} = \frac{15}{\quad}$$

$$\frac{2}{7} = \frac{\quad}{21} = \frac{10}{\quad} = \frac{\quad}{49}$$

$$\frac{5}{8} = \frac{15}{\quad} = \frac{\quad}{40} = \frac{25}{\quad}$$

c) Verifica daca sunt egale (prin produse incrucisate):

$$\frac{5}{8} \text{ si } \frac{15}{24} : \frac{\quad}{\quad} \times \frac{\quad}{\quad} = \frac{\quad}{\quad} \text{ si } \frac{\quad}{\quad} \times \frac{\quad}{\quad} = \frac{\quad}{\quad} \Rightarrow \text{-----}$$

$$\frac{7}{11} \text{ si } \frac{21}{34} : \frac{\quad}{\quad} \times \frac{\quad}{\quad} = \frac{\quad}{\quad} \text{ si } \frac{\quad}{\quad} \times \frac{\quad}{\quad} = \frac{\quad}{\quad} \Rightarrow \text{-----}$$

Exercitii: Capitol IV - Numere rationale (continuare)

Ex. 3. Fractii zecimale - transformari:

(10p)

a) Transforma in fractie ordinara ireductibila:

$0,6 = \frac{\quad}{\quad}$

$0,75 = \frac{\quad}{\quad}$

$0,125 = \frac{\quad}{\quad}$

$1,4 = \frac{\quad}{\quad}$

$2,25 = \frac{\quad}{\quad}$

$0,375 = \frac{\quad}{\quad}$

$0,04 = \frac{\quad}{\quad}$

$3,5 = \frac{\quad}{\quad}$

$0,625 = \frac{\quad}{\quad}$

b) Transforma in fractie zecimala (imparte numaratorul la numitor):

$\frac{3}{4} = \frac{\quad}{\quad}$

$\frac{7}{8} = \frac{\quad}{\quad}$

$\frac{1}{5} = \frac{\quad}{\quad}$

$\frac{5}{4} = \frac{\quad}{\quad}$

$\frac{9}{25} = \frac{\quad}{\quad}$

$\frac{11}{20} = \frac{\quad}{\quad}$

c) Scrie primele 5 zecimale ale fractiei $\frac{1}{7} = 0, (142857)$.

$\frac{1}{7} = 0, \text{-----} ; \frac{2}{7} = 0, \text{-----} ; \frac{3}{7} = 0, \text{-----}$

Ex. 4. Comparare si ordonare:

(10p)

a) Pune semnul corect < sau > (arata metoda folosita):

$\frac{3}{5} \text{ --- } \frac{4}{7}$

metoda: -----

$\frac{5}{8} \text{ --- } \frac{7}{12}$

metoda: -----

$\frac{4}{9} \text{ --- } \frac{5}{11}$

metoda: -----

$0,7 \text{ --- } \frac{3}{4}$

metoda: -----

$\frac{11}{15} \text{ --- } \frac{7}{10}$

metoda: -----

b) Ordoneaza CRESCATOR:

$\frac{5}{6}, \frac{3}{4}, \frac{7}{12}, \frac{2}{3}, \frac{11}{12}$

$\text{Numitor comun: c.m.m.m.c.}(6,4,12,3,12) = \text{---}$

$\text{Fractiile: ---/---, ---/---, ---/---, ---/---, ---/---}$

$\text{Ordine: ---} < \text{---} < \text{---} < \text{---} < \text{---}$

c) Ordoneaza DESCRESCATOR:

$0,75; \frac{5}{8}; 0,6; \frac{7}{10}; \frac{3}{5}$

$\text{Transformat in zecimale: ---, ---, ---, ---, ---}$

$\text{Ordine descrescatoare: ---} > \text{---} > \text{---} > \text{---} > \text{---}$

Exercitii: Capitol IV - Numere rationale (continuare)

Ex. 5. Adunare si scadere cu fractii ordinare:

(15p)

a) Calculeaza (simplifica rezultatul!):

$$1/3 + 1/4 = ______$$

$$5/6 - 1/4 = ______$$

$$3/8 + 5/12 = ______$$

$$7/10 - 3/5 = ______$$

$$2/3 + 3/4 = ______$$

$$5/6 + 7/9 = ______$$

$$11/12 - 3/8 = ______$$

$$3/5 + 7/15 - 2/3 = ______$$

b) Calculeaza pas cu pas:

$$1/2 + 1/3 + 1/6 =$$

$$c.m.m.m.c. = ______ \Rightarrow ______ / ______ + / - ______ / ______ = ______ / ______ = ______$$

$$3/4 - 1/6 + 5/12 =$$

$$c.m.m.m.c. = ______ \Rightarrow ______ / ______ + / - ______ / ______ = ______ / ______ = ______$$

$$7/8 + 5/6 - 11/12 =$$

$$c.m.m.m.c. = ______ \Rightarrow ______ / ______ + / - ______ / ______ = ______ / ______ = ______$$

Ex. 6. Inmultire si impartire cu fractii:

(15p)

a) Calculeaza (simplifica INAINTE daca posibil):

$$2/3 \times 3/4 = ______$$

$$5/8 \times 4/15 = ______$$

$$7/12 \times 9/14 = ______$$

$$3/5 \times 25 = ______$$

$$4/9 \times 27/16 = ______$$

$$15/8 \times 4/5 = ______$$

$$3/4 \div 1/2 = ______$$

$$5/6 \div 5/3 = ______$$

$$7/8 \div 7 = ______$$

$$9 \div 3/4 = ______$$

$$4/5 \div 8/15 = ______$$

$$14/9 \div 7/3 = ______$$

b) Numere mixte - transforma si calculeaza:

$$2 \frac{3}{4} = ______ / ______$$

$$5 \frac{1}{3} = ______ / ______$$

$$3 \frac{5}{8} = ______ / ______$$

$$22/7 = ______ ______ / ______$$

$$17/4 = ______ ______ / ______$$

$$41/9 = ______ ______ / ______$$

Exercitii: Capitol IV - Numere rationale (continuare)

Ex. 7. Operatii cu fractii zecimale:

(10p)

a) Adunare si scadere (aliniata virgulele!):

$$3,45 + 2,8 = ______$$

$$7,3 - 4,75 = ______$$

$$0,125 + 3,75 = ______$$

$$10,5 - 3,125 = ______$$

$$4,7 + 2,35 + 1,008 = ______$$

b) Inmultire si impartire:

$$1,5 \times 2,4 = ______$$

$$0,25 \times 0,4 = ______$$

$$3,6 \times 100 = ______$$

$$0,075 \times 1000 = ______$$

$$7,2 \div 0,9 = ______$$

$$4,5 \div 0,15 = ______$$

$$0,144 \div 1,2 = ______$$

$$3,75 \div 100 = ______$$

c) Rotunjeste la zecimi, sutimi si la intreg:

$$3,756: \quad \text{zecimi} = ______$$

$$\text{sutimi} = ______$$

$$\text{intreg} = ______$$

$$8,449: \quad \text{zecimi} = ______$$

$$\text{sutimi} = ______$$

$$\text{intreg} = ______$$

$$12,965: \quad \text{zecimi} = ______$$

$$\text{sutimi} = ______$$

$$\text{intreg} = ______$$

Ex. 8. Calculeaza respectand ordinea operatiilor:

(10p)

a) $1/2 \times (3/4 + 1/6) =$

Rezolvare:

b) $3/4 \div (1/2 - 1/3) + 5/6 =$

Rezolvare:

c) $(2/3 + 1/4) \times (5/6 - 1/3) =$

Rezolvare:

Exercitii: Capitol IV - Numere rationale (continuare)

Ex. 9. Probleme aplicative:

(10p)

a) Maria a citit $\frac{2}{5}$ dintr-o carte luni si $\frac{3}{8}$ marti.

Ce fractie din carte a citit in total? Ce mai are de citit?

Rezolvare:

Total citit = $\frac{\quad}{\quad}$; Ramas = $\frac{\quad}{\quad}$

b) Un rezervor contine $\frac{3}{4}$ din capacitate. Se adauga 120 litri,

ajungand la $\frac{5}{6}$ din capacitate. Cat este capacitatea totala?

Rezolvare:

Capacitate = $\quad$ litri

c) Un produs costa 45,60 lei. Se aplica o reducere de $\frac{1}{5}$.

Cat se plateste? Daca mai aplic o reducere de 0,1 din noul pret?

Rezolvare:

Dupa prima reducere = $\quad$ lei ; Dupa a doua = $\quad$ lei

d) Intr-o clasa, $\frac{3}{8}$ din elevi au luat nota 10, $\frac{1}{4}$ au luat 9,

iar restul au luat alte note. Ce fractie are alte note?

Daca sunt 32 de elevi, cati au luat 10? Cati au luat 9?

Rezolvare:

BONUS (10 puncte)

Ex. B1. Calculeaza (simplifica la maximum):

(4p)

$$(3/4 - 1/6) \div (5/8 + 7/12) \times (2/3 + 1/4) =$$

Ex. B2. Gaseste x rational:

(3p)

$$x + 3/4 = 5/6 \times (x - 1/2)$$

Ex. B3. Demonstreaza ca suma:

(3p)

$$1/1 \times 2 + 1/2 \times 3 + 1/3 \times 4 + \dots + 1/9 \times 10 = 9/10$$

(Indicatie: $1/(n(n+1)) = 1/n - 1/(n+1) \Rightarrow$ suma telescopica)

Barem de corectare

Ex. 1 10p 1p pentru fiecare (10 x 1p)

Ex. 2 10p a) 6x1p b) 3x1p + 0.5p fiecare blank c) 2p

Ex. 3 10p a) 9x1p b) 6x0.5p c) 1p

Ex. 4 10p a) 5x1p b) 3p c) 3p